KENYA MEDICAL TRAINING COLLEGE
FACULTY OF INFORMATION TECHNOLOGY AND MEDICAL EDUCATION
DEPARTMENT OF HEALTH RECORDS AND INFORMATION

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

PAPER: HEALTH STATISTICS

DATE: 27th SEPTEMBER, 2021 **TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the EXAMINATION **NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION ONE: MULTIPLE CHOICE QUESTIONS

1. Any numerical value calculated from sample data is called?

A. Error
B. Statistic
C. Bias
D. Mean

2. The variability which is defined as the difference between third and first quartile is considered as?

A. Quartile range
B. Deciles range
C. Percentile range
D. Interquartile range

3. The curve of cumulative frequency is also known as?

A. Ogive
B. Frequency polygon
C. A-give
D. Frequency curve

4. Snowball sampling is used for data collection through..... ?

A. Connecting relations
B. Small population
C. Representatives
D. None of the above

5. The list of all units in a population is called?

A. Random sampling
B. Sampling Frame
C. Bias
D. Parameter

6. Which of the following symbols denotes the population mean?

A. σ
B. $\bar{x}$
C. μ
D. $\sigma 2$

7. _____ are those which are collected fresh and are original in character

A. Primary data
B. Secondary data
C. Qualitative data
D. Quantitative data

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8. In a symmetrical distribution, the sum of the positive and negative deviations from mean, mode, median is?

A. 0
B. 1
C. -1
D. 2

9. A distribution with two modes is known as?

a. Double modal Distribution
b. Bi-modal distribution
c. Multimodal distribution
d. Unimodal distribution

10. The range of the scores 20, 4, 145, 27, 199 is?

A. 140
B. 141
C. 195
D. 7

11. ..... is the name given to the square of standard deviation

A. Mean deviation
B. Coefficient of variation
C. Variance
D. None of the above

12. _____ indicates a plan of action to be carried out in connection with a proposed

A. Research work.
B. Research proposal
C. Research abstract
D. Research methodology

13. The measurement scale which allows the ranking of numbers rather than arithmetic operations on data is classified as?

A. Valid scale
B. Continuous scale
C. Discrete scale
D. Ordinal scale

14. The data table which is presented in tabular form on the basis of single characteristics is classified as?

A. Interval table
B. Complex table
C. Percentage table
D. Simple table

15. The type of cumulative frequency distribution in which the class intervals are added in bottom to top order is classified as?

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A. More than type distribution
B. Marginal distribution
C. Variation distribution
D. Less than type distribution

16. If the quartile range is 48 then the Quartile deviation is?
A. 12
B. 24
C. 36
D. 72

17. Sum of the deviations of a variable from its mean is always?
A. 0
B. 1
C. 2
D. 5

18. The most frequent occurring observation in the data is called?
A. mean
B. median
C. mode
D. standard deviation

19. For a variable X, the nth positive root of the product of the x1, x2, x3,.... Xn is called?
A. Arithmetic mean
B. harmonic mean
C. standard mean
D. geometric mean

20. The following are measure of central tendency which one is NOT?
A. Range
B. Mean
C. Mode
D. Median.

21. The following are all methods of data collection except?
A. Observation
B. Sampling
C. Abstraction
D. Questionnaire

22. The procedure by which we assign numbers to opinions, attitude and concepts is called?
A. Coding
B. Scaling
C. Classification
D. None of the above

23. The following are the Relative measures of dispersion except?
A. Co-efficient of Mean deviation
B. Standard deviation
C. Co-efficient of Range
D. Co-efficient of Quartile deviation

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24. The largest value is 60 and the smallest value is 40 and the number of classes desired is 5 then the class interval is?

A. 20
B. 4
C. 25
D. 15

25. The classification on the basis of time order is called?

A. Disclosed classification
B. Chronological classification
C. External classification
D. Internal classification

26. If the midpoints of bars on the charts are marked and the marked dots are joined by a straight line then this graph is classified as?

A. Frequency curve
B. Cumulative frequency curve
C. Frequency polygon
D. Histogram

27. The government publications are considered as?

A. Primary source
B. Secondary source
C. Primary and secondary source
D. None of the above

28. The type of questions included in questionnaire to record responses in which the respondent can answer in any way are classified as?

A. Multiple choices
B. Itemized questions
C. Open ended questions
D. Closed ended questions

29. The average number of patients expected to have been treated on a bed (used a bed) in a specified period?

A. Turnover per bed
B. Occupied bed days
C. Turnover interval
D. Average length of stay

30. The total area under the normal curve is equal to?

A. 100
B. 0
C. 1
D. 2

31. In ..... correlation the points lie close to a straight line, with y decreasing as x increases.

A. Negative correlation
B. Positive correlation
C. No correlation
D. None of the above

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32. Which of the following is not a probability distribution?

A. Poisson Distribution
B. Normal distribution
C. Systematic Distribution
D. Binomial Distribution

33. The basic graphical tool that illustrates the relationship between 2 variables is known as?

A. Correlation
B. Scatter diagrams
C. Histogram
D. Pie chart

34. The process of predicting variable Y using Variable X is known as?

A. Skewness
B. Scatter plots
C. Regression
D. Correlation

35. The key methods of data collection are the following except?

A. Abstraction
B. Sampling
C. Observation
D. Interview

36. The shape of the Normal Curve is _____?

A. Bell Shaped
B. Flat
C. Circular
D. Spiked

37. What do statisticians call birth, death, marriage, and divorce?

A. Vital statistics
B. Census categories
C. Sample surveys
D. General records

38. A statement made about a population for testing purpose is called?

A. Statistic
B. Hypothesis
C. Level of Significance
D. Test-Statistic

39. The formal process whereby a person is accepted by a hospital for the purpose of hospital treatment as an inpatient is known as?

A. Admission
B. Transfer in
C. Abscondee
D. Referral

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40. .....Is the proportion of deaths within a designated population of cases over the course of the disease

A. Incidence
B. Prevalence
C. Case fatality rate
D. Crude death Rate

SECTION TWO: SHORT ANSWER QUESTIONS (40 MARKS)

1. Differentiate between the two types of hypothesis and give an example in each case(5 marks)
2. Use the data below for a certain sample to calculate the given measures of central tendency and dispersion (5 marks)

   8, 12, 23, 28, 22, 26, 30, 35, 42

   a) Mean (2 marks)
   b) Standard Deviation(3 marks)

3. Briefly explain the various types of correlation (5 marks)
4. List any 5 sources of Literature review
5. Briefly explain any 5 methods of Data Presentation
6. State any 5 measures of mortality rates (5 marks)
7. Using a period of 25 days, allocated bed of 35 and OBD of 900. Average length of stay of 20 days. Calculate (10 marks)
   a. Percentage occupancy (2 marks)
   b. Discharges and Deaths (2 marks)
   c. Turn over per Bed (2 marks)
   d. Explain your results in a and c above (4 marks)

SECTION 3: LONG ANSWER QUESTIONS (20 marks)

1. Test whether there is a significant association between smoking and Lung cancer at 95% confidence level. (10 marks)

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		Lung Cancer	
		YES	NO
SMOKING	YES	39	21
	NO	29	19

2. The finance division of a large company is investigating its procedures for the selection of new accountancy trainees. Potential applicants are given prior to appointment both a written test and a formal interview. The performances of eight successful applicants were rated after their first full year with the company. The independent rankings of written test, interview assessment and job performance for the eight trainees are given as follows (10 marks)

[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
Written Test	6	2	7	4	1	5	3	8
Interview assessment	1	4	2	3	6	5	8	7
Job performance	1	2	3	4	5	6	7	8

Calculate a rank correlation coefficient between

a) Job performance and written test (4 marks)

b) Job performance and interview assessment (4 marks)

c) Interpret your results of a and b above (2 marks)

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	Probability level (alpha)					
df	0.5	0.10	0.05	0.02	0.01	0.001
1	0.455	2.706	3.841	5.412	6.635	10.827
2	1.386	4.605	5.991	7.824	9.210	13.815
3	2.366	6.251	7.815	9.837	11.345	16.268
4	3.357	7.779	9.488	11.668	13.277	18.465
5	4.351	9.236	11.070	13.388	15.086	20.517

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